

Tutorial 4

1. A base station has 5 channels for two groups of users to access. The users in both groups generate traffic requests randomly, each of which requires one channel, following the Poisson distribution. The users in the first group generate requests with mean 1.0 request per time period, while the users in the second group generate requests with mean 3.0 requests per time period. In one period, if there are more requests than the available channels, the exceeded requests will be blocked regardless of their associated group of users. Calculate the average number of occupied channels of the base station per time period. For the Poisson process, the probability is obtained as follows:

$$P(A=k) = \exp(-m T) * ((m T)^k) / k!$$

2. Consider a circle service area with a base station, user A and user B. The base station is located in the middle of the circle service area while user A is fixed and located 3 km away from the base station (Figure Q2). The radius of the circle service area is 10 km. The transmit power of user A is 10dB and the transmit power of user B is 15dB. Assume that the base station can be captured by one of the users if the receive power in watt of that user at the base station is at least two times higher than that of the other user. Assume that user B can move in the service area randomly and uniformly. The receive power in watt is given by $P_R = P_T / ((1+D)^E)$, where P_T is the transmit power in watt, D is the distance between the base station and the user in kilometer, and $E=2$.

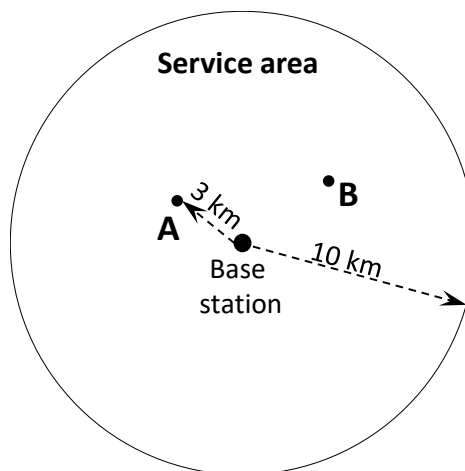


Figure Q2

- (i) Calculate the probability that user A captures the base station.
- (ii) Calculate the probability that user B captures the base station.
- (iii) Calculate the probability that none of the users captures the base station

3. Consider a packet transmission from a sender to its receiver. If the sender has a packet to transmit, it will transmit immediately in a time slot. The packet transmission is successful with probability 0.3. If the packet transmission is unsuccessful, the sender applies the binary exponential backoff mechanism. In particular, if the packet transmission is not successful, the sender will choose a random integer number X between 1 and $BO=10$. Then the sender waits for X time slots before retransmission. If the retransmission is still unsuccessful, BO will be doubled, and the same procedure is repeated. If the retransmission is successful, BO will be reset to the initial value (i.e., $BO=10$). The retransmission of the same packet can be repeated for 5 times excluding the first transmission without backoff. That is, if the same packet cannot be successfully transmitted within 6 tries, it will be discarded. Assume that the packet transmission takes one time slot. Ignore the time that is taken for ACK.

- (i) Calculate the probability that a packet will be successfully transmitted within 6 tries.
- (ii) Calculate the average delay only for the packets that are successfully transmitted by the sender.